# Markdown and webpages with knitr

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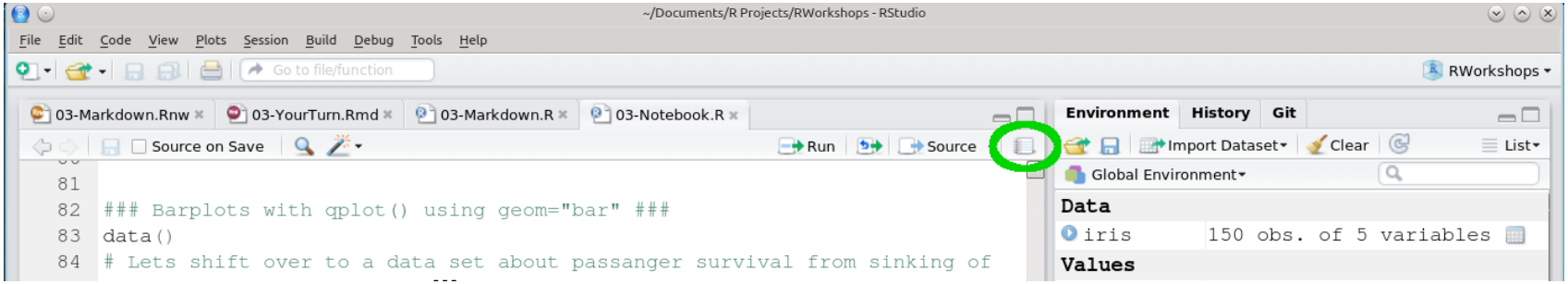
# Knitr can also be used to create webpages

- Compile “notebooks" displaying your code and graphics in a single webpage
- Use knitr to create multi-purpose webpages
- R presentations with knitr and RStudio
- `knitrBootstrap` for more advanced HTML/Markdown options

# R Code Notebooks

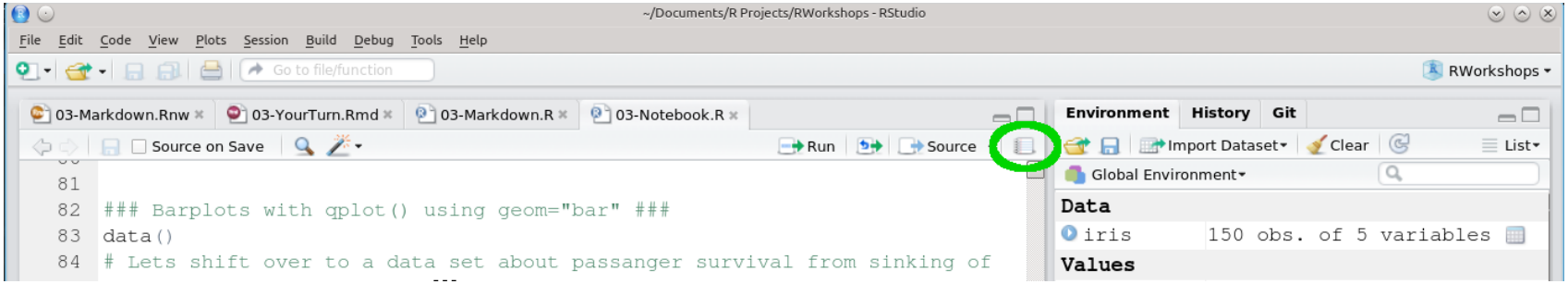
Rstudio will compile an R script into an HTML notebook that includes all generated pictures, output, and code.

This may make it easier for you to remember what a specific code file does and easily present your analyses.



# Your Turn

Open the file 03-Notebook.R and compile it into an R notebook. It may take a few moments to compile.



# What is markdown?

Markdown is a text-based format that can be easily converted into HTML.

```
Header 1
-------------------------------

*italics* and **bold** text

Lists are easy too:
- Item 1
- Item 2
- Item 3
```

# R markdown

R markdown is very similar to the LaTeX code chunks we used earlier.
To create a code chunk (in any language), use

```
```
This is a code chunk
It will have fixed-width font
```
```

To create an R code chunk,

```
```{r chunkname, fig.width=7, fig.height=6}
x <- rnorm(100)
library(ggplot2)
qplot(x, geom="histogram")
```
```

# Inline R code in Markdown

To make inline R code in markdown, use ‘`r rnorm(20)`‘

This is analogous to `\Sexpr{}` in LaTeX documents

Protip: Click on the “?” button in the toolbar on a .Rmd document in Rstudio to get a Markdown cheat-sheet!

# Your Turn

- In RStudio, create a new R Markdown File (File $->$ New File $->$ R Markdown)
- Change the title of the document, add a couple of section headers, try out bold and italic text, make a table, and add an R code chunk
- In your R code chunk, save 100 randomly generated x values and plot a histogram
- Outside your code chunk, calculate the mean of the values you generated and reference that using an inline code chunk
- Compile your document

# Equations in R Markdown

R Markdown also includes MathJax equations, which allow LaTeX equation syntax in web pages.
Single-line equations:

```
$x=y=z$
```

Multi-line equations:

```
$$
a x^2 + bx + c = 0\\
... = 0\\
x = \frac{1}{2a}\left(-b \pm \sqrt{b^2 - 4ac}\right)
$$
```

Some rules apply

# Tables and Markdown

`knitr` can be used to make HTML tables with the `kable` function:

```
library(knitr)
kable(dataframe)
```

`stargazer` will also create HTML tables for regression output:

```
stargazer(lm(y~x, data=dataframe), type="html")
```

# Your Turn

- Print the first 6 lines of the iris data to a HTML table in a markdown document
  Don't forget `results='asis'` in the code chunk definition!
- Fit a linear regression using Petal Length to describe Sepal Length
- Output the results of the linear regression to an HTML table using stargazer

# Figures and Images in Markdown

Knitr will automatically include your images in markdown pages just as it did in LaTeX, but sometimes it's easier to manually include an image yourself.

```
![Alt text](/path/to/img.jpg)
```

```
![Alt text](/path/to/img.jpg "Optional title")
```

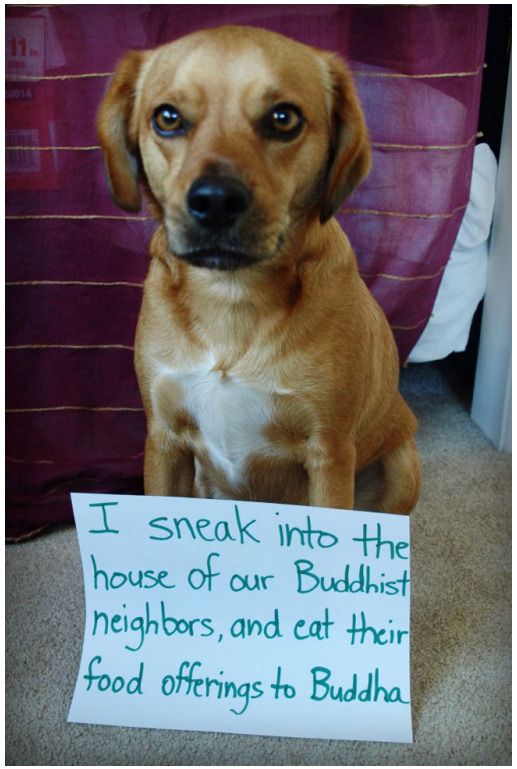
If you want to change the dimensions of the image, you may need to use HTML:

```
<img src="path/to/img.jpg" alt="some text"
      width="42" height="42">
```

# Your Turn

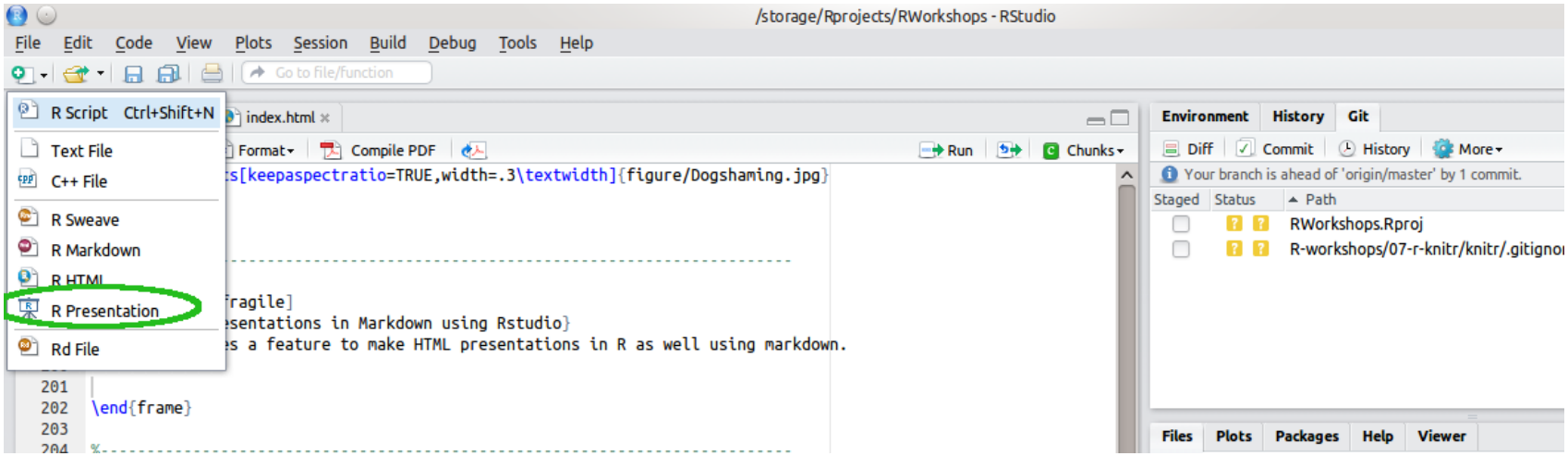
Practice with images by inserting an image into the markdown document manually using both markdown syntax and HTML syntax.

If you don't have a favorite image, try this one



# Presentations in Markdown using Rstudio

RStudio provides a feature to make HTML presentations in R as well using markdown.



# Presentations in Markdown using Rstudio

These HTML presentations are created using similar markdown-style syntax

```
Title
=========================================
slide contents
* item 1
* item 2
```

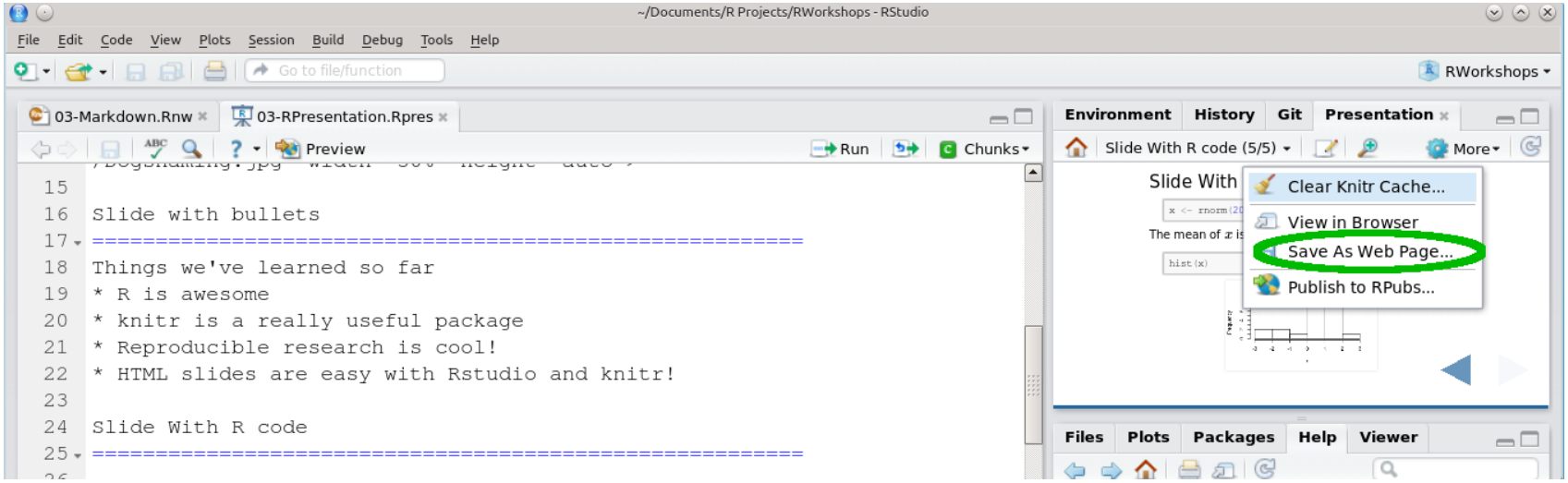
# Your Turn

Try some of the following:

- Create a new R presentation
- Change the title, the author, the date
- On the first slide, include a picture using markdown syntax
- On the next slide, include a picture using HTML syntax: `<img src="path/to/image" width='50%' height='auto'>`
- On the third slide, make a bulleted list
- On the fourth slide, generate 20 random numbers in an R code chunk and compute the mean. If you're feeling brave, add a histogram on the same slide.

# Saving R presentations

Rstudio creates a new tab to “preview" your presentation, but you have to manually save it as an HTML page.




You could also just present within RStudio, but it may be advantageous to have the HTML file for portability.

# Extras

If you have the Preview Release of Rstudio, you can also use the `knitrBootstrap` package to theme your knitr markdown documents. The file 03-MarkdownThemes.R provides some code if you want to try it out.

If you get bored with the default presentation theme, you can try customizing slide themes.

These features are relatively new, but can make even cooler looking webpages, and allows you much finer control over the appearance of the final HTML output.

# Questions

Anyone have any questions?

Please take the time to fill out the survey and let us know what we could do better!